Control for CUSF

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1 Modelling

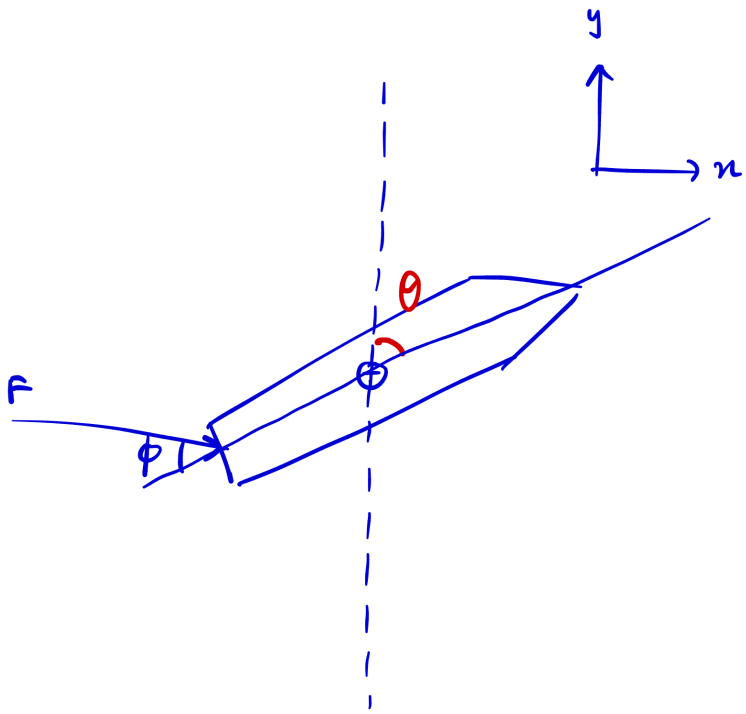


Figure 1: 3DOF Thrust Vector Controlled Rocket

1. Consider a 3DOF thrust vector controlled rocket, shown in figure 1. Write the equations of motion.
2. Write the equations of motion in nonlinear state space form.
3. Write a function in MATLAB to implement this differential equation. The inputs should be the state $(x, y, \theta, \dot{x}, \dot{y}, \dot{\theta})$ and the inputs (F, ϕ).
4. For suitable initial conditions, try to predict what would happen analytically/intuitively and verify your simulation.

2 Controller Design

1. Using the 'jacobian(f,x)' function in MATLAB, find the Jacobian of the nonlinear differential equation with respect to $\mathbf{x} = \left[x, y, \theta, \dot{x}, \dot{y}, \dot{\theta}\right]^\top$ and input $\mathbf{u} = [F, \phi]$.
2. Substitute in the right parameters (there's a function for this too) and compute the 'A' and 'B' matrices to form a linear state space equation.
3. Define your state space system.
4. Compute the controllability matrix, using the 'ctrb(A, B)' function. Check if the row rank condition is satisfied to verify that the system is reachable/controllable locally.
5. Identify where you might want your poles - this may take some trial and error, and use the 'place(A, B, p)' function to compute the gain matrix K for $\mathbf{u} = -\mathbf{K}\mathbf{x}$.
6. Create a closed loop Simulink block diagram for this and run the simulation.